Randomized counting-to-decision reduction for exponent-p class-two matrix-group isomorphism

Joshua A. Grochow, Youming Qiao

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Editorial scope. Counted once: randomized group counting-to-decision Theorem B and its full B-prime isometry mechanism. All four gadget requirements, verification, functionality, projective orbit counting, random-subspace/classical-group probability analysis and group multiplicity calculation retained. Preserves all original prime/field/exponent/class/input and randomness conditions; no support results or parameter regimes counted separately.

Allowed standard prerequisites (editorial).

- Finite-field subspace counting, matrix-algebra/module algorithms and random generation of classical groups. Complete source-local Section5 proof retained, including Claims 19 and20 and Proposition15 probability analysis. Exact imported Theorem21 on generation of SL(n,q) and constructive Baer group transfer retained with the complete74-entry bibliography. Imported classical group and module algorithms are not recursively reproved.

Author notation and isometry models

Vector spaces. Let $\mathbb{F}$ be a field. In this paper we only consider finite-dimensional vector spaces over $\mathbb{F}$. We use $\mathbb{F}^n$ to denote the vector space of length-n column vectors. The ith standard basis vector of $\mathbb{F}^n$ is denoted $\vec{e}_i$. Depending on the context, 0 may denote the zero vector space, a zero vector, or an all-zero matrix. For S a set of vectors, we use $\langle S\rangle$ to denote the subspace spanned by elements in S.

Some groups. The general linear group of degree n over a field $\mathbb{F}$ is denoted by $\mathrm{GL}(n,\mathbb{F})$. The symmetric group of degree n is denoted by S_n. The natural embedding of S_n into $\mathrm{GL}(n,\mathbb{F})$ is to represent permutations by permutation matrices. The subgroup of $\mathrm{GL}(n,\mathbb{F})$ consisting of diagonal matrices is called the diagonal subgroup, denoted by $\mathrm{diag}(n,\mathbb{F})$. A monomial matrix is a product of a diagonal and a permutation matrix; equivalently, each row and each column has exactly one non-zero entry. The collection of monomial matrices forms a subgroup of $\mathrm{GL}(n,\mathbb{F})$, which we call the monomial subgroup and denote by $\mathrm{Mon}(n,\mathbb{F})$. It is the semi-direct product $\mathrm{diag}(n,\mathbb{F}) \rtimes S_n \cong (\mathbb{F}^*)^n \rtimes S_n$.

Matrices. Let $\mathrm{M}(\ell \times n, \mathbb{F})$ be the linear space of $\ell \times n$ matrices over $\mathbb{F}$, and $\mathrm{M}(n, \mathbb{F}) := \mathrm{M}(n \times n, \mathbb{F})$. Given $A \in \mathrm{M}(\ell \times n, \mathbb{F})$, A^t denotes the transpose of A. A matrix $A \in \mathrm{M}(n, \mathbb{F})$ is alternating, if for any $u \in \mathbb{F}^n$, $u^t A u = 0$. That is, A represents an alternating bilinear form. Note that in characteristic $\neq 2$, alternating is the same as skew-symmetric, but in characteristic 2 they differ (in characteristic 2, skew-symmetric=symmetric). The linear space of $n \times n$ alternating matrices over $\mathbb{F}$ is denoted by $\Lambda(n, \mathbb{F})$. The $n \times n$ identity matrix is denoted by I_n, and when n is clear from the context, we may just write I. The elementary matrix $E_{i,j}$ is the matrix with the (i,j)th entry being 1, and other entries being 0. The (i,j)-th elementary alternating matrix is the matrix $E_{i,j} - E_{j,i}$.

Matrix tuples. We use $\mathrm{M}(\ell \times n, \mathbb{F})^m$ to denote the linear space of m-tuples of $\ell \times n$ matrices. Boldface letters like $\mathbf{A}$ and $\mathbf{B}$ denote matrix tuples. Let $\mathbf{A} = (A_1, \ldots, A_m)$, $\mathbf{B} = (B_1, \ldots, B_m) \in \mathrm{M}(\ell \times n, \mathbb{F})^m$. Given $P \in \mathrm{M}(\ell, \mathbb{F})$ and $Q \in \mathrm{M}(n, \mathbb{F})$, $P\mathbf{A}Q := (PA_1Q, \ldots, PA_mQ) \in \mathrm{M}(\ell, \mathbb{F})$. Given $R = (r_{i,j})_{i,j \in [m]} \in \mathrm{M}(m, \mathbb{F})$, $\mathbf{A}^R := (A'_1, \ldots, A'_m) \in \mathrm{M}(m, \mathbb{F})$ where $A'_i = \sum_{j \in [m]} r_{j,i} A_j$.

Remark5. In particular, note that the coefficients in the formula of defining A'_i correspond to the entries in the ith column of R. While this choice is immaterial (we could have chosen the opposite convention), all of our later calculations are consistent with this convention.

Given $\mathbf{A}, \mathbf{B} \in \mathrm{M}(\ell \times n, \mathbb{F})^m$, we say that $\mathbf{A}$ and $\mathbf{B}$ are isometric, if there exists $P \in \mathrm{GL}(n, \mathbb{F})$, such that $P^t \mathbf{A} P = \mathbf{B}$. Finally, $\mathbf{A}$ and $\mathbf{B}$ are pseudo-isometric if there exist $P \in \mathrm{GL}(n, \mathbb{F})$ and $R \in \mathrm{GL}(m, \mathbb{F})$, such that $P^t \mathbf{A} P = \mathbf{B}^R$.

Matrix spaces. Linear subspaces of $\mathrm{M}(\ell \times n, \mathbb{F})$ are called matrix spaces. Calligraphic letters like $\mathcal{A}$ and $\mathcal{B}$ denote matrix spaces. By a slight abuse of notation, for $\mathbf{A} \in \mathrm{M}(\ell \times n, \mathbb{F})^m$, we use $\langle \mathbf{A} \rangle$ to denote the subspace spanned by those matrices in $\mathbf{A}$. For $\mathbf{A}, \mathbf{B} \in \mathrm{M}(n, \mathbb{F})^m$, we say that the spaces $\langle \mathbf{A} \rangle, \langle \mathbf{B} \rangle$ are isometric iff the tuples $\mathbf{A}, \mathbf{B}$ are pseudo-isometric.

Definition4 (The Alternating Matrix Space Isometry problem). Input: $A_1, \ldots, A_m$ and $B_1, \ldots, B_m$, $n \times n$ alternating matrices over a field $\mathbb{F}$. Decide: Is there a $L \in \mathrm{GL}(n, \mathbb{F})$, such that the linear span of $\{A_i : i \in [m]\}$ is equal to the linear span of $\{L^t B_i L : i \in [m]\}$?

3-way arrays. Let $\mathrm{T}(\ell \times n \times m, \mathbb{F})$ be the linear space of $\ell \times n \times m$ 3-way arrays over $\mathbb{F}$. We use the fixed-width teletypefont for 3-way arrays, like $\mathtt{A}, \mathtt{B}$, etc.. Given $\mathtt{A} \in \mathrm{T}(\ell \times n \times m, \mathbb{F})$, we can think of $\mathtt{A}$ as a 3-dimensional table, where the (i,j,k)th entry is denoted as $\mathtt{A}(i,j,k) \in \mathbb{F}$. We can slice $\mathtt{A}$ along one direction and obtain several matrices, which are then called slices. For example, slicing along the first coordinate, we obtain the horizontal slices, namely ℓ matrices $A_1, \ldots, A_\ell \in \mathrm{M}(n \times m, \mathbb{F})$, where $A_i(j,k) = \mathtt{A}(i,j,k)$. Similarly, we also obtain the lateral slices by slicing along the second coordinate, and the frontal slices by slicing along the third coordinate. We will often represent a 3-way array as a matrix whose entries are vectors. That is, given $\mathtt{A} \in \mathrm{T}(\ell \times n \times m, \mathbb{F})$, we can

write

$$\mathtt{A} = \begin{bmatrix} w_{1,1} & w_{1,2} & \cdots & w_{1,n} \\ w_{2,1} & w_{2,2} & \cdots & w_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{\ell,1} & w_{\ell,2} & \cdots & w_{\ell,n} \end{bmatrix},$$

where $w_{i,j} \in \mathbb{F}^m$, so that $w_{i,j}(k) = \mathtt{A}(i,j,k)$. Note that, while $w_{i,j} \in \mathbb{F}^m$ are column vectors, in the above representation of $\mathtt{A}$, we should think of them as along the direction "orthogonal to the paper." Following [45], we call $w_{i,j}$ the tube fibers of $\mathtt{A}$. Similarly, we can have the row fibers $v_{i,k} \in \mathbb{F}^n$ such that $v_{i,k}(j) = \mathtt{A}(i,j,k)$, and the column fibers $u_{j,k} \in \mathbb{F}^\ell$ such that $u_{j,k}(i) = \mathtt{A}(i,j,k)$. Given $P \in \mathrm{M}(\ell, \mathbb{F})$ and $Q \in \mathrm{M}(n, \mathbb{F})$, let $P\mathtt{A}Q$ be the $\ell \times n \times m$ 3-way array whose kth frontal slice is PA_kQ. For $R = (r_{i,j}) \in \mathrm{GL}(m, \mathbb{F})$, let $\mathtt{A}^R$ be the $\ell \times n \times m$ 3-way array whose kth frontal slice is $\sum_{k' \in [m]} r_{k',k} A_{k'}$. Note that these notations are consistent with the notations for matrix tuples above, when we consider the matrix tuple $\mathbf{A} = (A_1, \ldots, A_k)$ of frontal slices of $\mathtt{A}$.

TheoremB. For p an odd prime, $p \geq n^{\Omega(1)}$, there is a randomized counting-to-decision reduction for GpIso2Exp(p) for groups of order p^n, running in time $p^{O(n)} = \mathrm{poly}(|G|)$.

TheoremB'. For q a prime power with $q = n^{\Omega(1)}$, there is a randomized counting-to-decision reduction for Alternating Matrix Space Isometry which, given $n \times n$ alternating matrix spaces $\mathcal{A}, \mathcal{B}$ over $\mathbb{F}_q$ of dimension m, computes the number of isometries from $\mathcal{A}$ to $\mathcal{B}$ in time $q^{O(n)}$. The reduction queries the decision oracle with inputs of dimension at most $O(n^2)$.

5 Counting-to-decision reduction by restricting to diagonal groups

In this section, we devise a gadget to achieve the restriction to the group of diagonal matrices, and use it to do the counting to decision reduction for Alternating Matrix Space Isometry.

5.1 Preliminaries

Some preparations are in order.

Observation12. Let $n \geq 23$. Then any permutation $\sigma \in S_n$ either fixes a set of 6 points $P \subseteq [n]$, or moves a set of 6 points $P \subseteq [n]$ to another set of 6 points $Q \subseteq [n]$ such that these two sets are disjoint.

Proof. Suppose σ fixes at most 5 points. Then there are at least 18 points that are not fixed by σ. Suppose σ has t non-trivial cycles of length $l_1, \ldots, l_t$, such that $\sum_i l_i \geq 18$. For a cycle $(p_1, \ldots, p_s)$, we can choose $p_1, p_3, \ldots, p_{2\lfloor s/2 \rfloor - 1}$ and put them in P, and $p_2, p_4, \ldots, p_{2\lfloor s/2 \rfloor}$ in Q. Do this for every cycle, we obtain the desired P and Q. The worst case is when every cycle is of length 3. Since there are at least 18 points not fixed by σ, P is of size ≥ 6. □

We shall make repeated uses of the following facts.

Fact13.

1. Given $a_i \in \mathbb{R}$, $0 \le a_i \le 1$, $i \in [m]$, $\prod_{i\in[m]}(1-a_i) \ge 1 - \sum_{i\in[m]} a_i$.
2. Let $m, N \in \mathbb{N}$ and $1 \le m \le N$. A random matrix $A \in \mathrm{M}(N \times m, q)$ is of rank m with probability $\ge 1 - 2/q^{N-m+1}$.
3. For $n \in \mathbb{N}$, $0 \le d \le n$, the number of dimension-d subspaces of $\mathbb{F}_q^n$ is equal to the Gaussian binomial coefficient
$$\binom{n}{d}_q := \frac{(q^n-1)\cdot(q^n-q)\cdots(q^n-q^{d-1})}{(q^d-1)\cdot(q^d-q)\cdots(q^d-q^{d-1})}.$$
4. The Gaussian binomial coefficient satisfies $q^{(n-d)d} \le \binom{n}{d}_q \le q^{(n-d)d+d}$.
5. For $d \in \mathbb{N}$, the number of complement subspaces of a fixed dimension-d subspace of $\mathbb{F}_q^n$ is $q^{d(n-d)}$.

Proof. For(2), $\Pr[\mathrm{rk}(A) = m] = (1-1/q^N)\cdot(1-q/q^N)\cdots(1-q^{m-1}/q^N)$. By(1), we have

$$\Pr[\mathrm{rk}(A) = m] \ge 1 - \sum_{i=N-m+1}^{N} \frac{1}{q^i} = 1 - \frac{1}{q^{N-m+1}} - \sum_{i=N-m+2}^{N} \frac{1}{q^i} \ge 1 - \frac{2}{q^{N-m+1}}.$$

□

5.2 Describing the gadget

Let $\mathcal{A} \le \Lambda(n,q)$ be an alternating matrix space, and let $\mathbf{A} = (A_1, \ldots, A_m) \in \Lambda(n,q)^m$ be an ordered linear basis of $\mathcal{A}$. Let $\mathtt{A} \in \mathrm{T}(n \times n \times m, \mathbb{F}_q)$ be the 3-way array constructed from $\mathbf{A}$, i.e. the ith frontal slice of $\mathtt{A}$ is A_i. We shall assume $n = \Omega(1)$, and $q = n^{\Omega(1)}$ throughout the remainder of this section.

The form of the gadget. To describe the gadget, it is easier to view $\mathtt{A}$ from the lateral viewpoint. That is, for $i \in [n]$, let $C_i = [A_1 e_i, \ldots, A_m e_i] \in \mathrm{M}(n \times m, q)$. Let $\mathbf{C} = (C_1, \ldots, C_n) \in \mathrm{M}(n \times m, q)^n$. Then construct $\mathbf{C}' = (C'_1, \ldots, C'_n)$, $C'_i = \begin{bmatrix} C_i & 0 \\ 0 & G_i \end{bmatrix}$, where G_i is of size $6n \times 4n^2$. For $i \in [n]$, $G_i = [0 \ \cdots \ 0 \ H_i \ 0 \ \cdots \ 0]$, where H_i is of size $6n \times 4n$ in the ith block, and 0 denotes an all-zero matrix of size $6n \times 4n$. The H_i will be described below. Note that from the frontal viewpoint of looking at $\mathtt{A}$, G_i's are inserted, vertically, below and behind $\mathtt{A}$. So to preserve the alternating structure, $-G_i$'s also need to be inserted, horizontally, on the right and behind $\mathtt{A}$. We therefore get $\widetilde{\mathtt{A}}$, which is of size $7n \times 7n \times (m + 4n^2)$.

Conditions imposed on the H_i's. Of course, the key to the construction above lies in the properties of the H_i's. Let $V_i \leq \mathbb{F}_q^{6n}$ be the subspace spanned by the columns of H_i. We shall impose the following conditions on H_i.

1. For any $i \in [n]$, $\mathrm{rk}(H_i) = \dim(V_i) = 4n$.
2. For any $i, j \in [n]$, $i \neq j$, $\mathrm{rk}([H_i\ H_j]) = \dim(V_i \cup V_j) = 6n$.
3. For any $(i_1, i_2, i_3, i_4, i_5, i_6) \in [n]^6$ and $(j_1, j_2, j_3, j_4, j_5, j_6) \in [n]^6$, such that $|\{i_1, \ldots, i_6\} \cup \{j_1, \ldots, j_6\}| = 12$, i.e. i_k and j_ℓ all different, the coset $\mathcal{C} = \{T \in \mathrm{GL}(6n, q) : \forall k \in [6], T(V_{i_k}) = V_{j_k}\}$ is empty. Note that for any $i \in [n]$, $T(V_i)$ is spanned by the columns of TH_i.
4. For any $(i_1, i_2, i_3, i_4, i_5, i_6) \in [n]^6$, i_k all different, the group $\mathcal{S} = \{T \in \mathrm{GL}(6n, q) : \forall k \in [6], T(V_{i_k}) = V_{i_k}\}$ consists of only of scalar matrices.

Remark14. Given $H_1, \ldots, H_n \in \mathrm{M}(6n \times 4n, q)$, whether they satisfy the four conditions can be verified in polynomial time.

Conditions(1) and(2) are easily verified in deterministic polynomial time. For condition(3), it can be formulated as a linear algebraic problem as follows. Let X be a $6n \times 6n$ variable matrix. Let Y_k, $k \in [6]$, be $4n \times 4n$ variable matrices. Set up the equations $XH_{i_k} = H_{j_k}Y_k$, and solve the linear equations to get a subspace of $\mathbb{F}_q^{(6n)^2+6\cdot(4n)^2}$. The question is then whether this subspace contains $(T, R_1, \ldots, R_6)$ where $T \in \mathrm{GL}(6n, q)$ and $R_i \in \mathrm{GL}(4n, q)$. This is an instance of the symbolic determinant identity testing (SDIT) problem, so it admits a randomized efficient algorithm when $q = n^{\Omega(1)}$. In fact, this instance of SDIT problem can be solved in deterministic polynomial time. For this let us also check out condition(4). Here, let X and Y_i be from above, and set up the equations $XH_{i_k} = H_{i_k}Y_k$. Solve the linear equations to get a subspace of $\mathbb{F}_q^{(6n)^2+6\cdot(4n)^2}$. This subspace turns out to be an algebra under the natural multiplications. Indeed, if $AH_{i_k} = H_{i_k}B_k$ and $A'H_{i_k} = H_{i_k}B'_k$, then $AA'H_{i_k} = H_{i_k}B_kB'_k$. To compute the unit group in a matrix algebra can be solved by a polynomial-time Las Vegas algorithm by [16]. Given the unit group, whether it consists of only scalar matrices can be verified easily in deterministic polynomial time. Then the linear space in condition(3) is a module over the algebra defined in the last paragraph. Because of this structure, the SDIT problem for such instances can be solved in deterministic polynomial time [14,19,37].

5.3 Construction and properties of the gadget

The following three propositions reveal the construction and functions of the gadget described above. First about the construction. Instead of constructing the above H_i's explicitly in a deterministic way, we shall show that random choices suffice.

Proposition15. Let $H_i \in \mathrm{M}(6n \times 4n, q)$, $i \in [n]$, be random matrices. Then H_i's satisfy the four conditions in Section5.2 with probability $\geq 1 - n^{O(1)}/q^{\Omega(1)}$.

Second about the functionality. The following proposition formally explains this.

Proposition16. Suppose $\mathtt{A}$ and $\mathtt{B}$ are two 3-tensors constructed from ordered bases of m-dimensional alternating matrix spaces $\mathcal{A}, \mathcal{B} \leq \Lambda(n, q)$. Let $\widetilde{\mathtt{A}}$ and $\widetilde{\mathtt{B}}$ be constructed as above, and let $\widetilde{\mathcal{A}}$ and $\widetilde{\mathcal{B}}$ be the alternating matrix spaces spanned by the frontal slices of $\widetilde{\mathtt{A}}$ and $\widetilde{\mathtt{B}}$, respectively. Then $\mathcal{A}$ and $\mathcal{B}$ are isometric via a diagonal matrix if and only if $\widetilde{\mathcal{A}}$ and $\widetilde{\mathcal{B}}$ are isometric.

Finally we shall use this gadget to achieve a counting-to-decision reduction for Alternating Matrix Space Isometry. Formally, we have the following.

Proposition17. Suppose we are given $\mathcal{A}, \mathcal{B} \leq \Lambda(n, q)$ and a decision oracle for Alternating Matrix Space Isometry. Then there exists a Las Vegas randomized algorithm that computes the number of isometries from $\mathcal{A}$ to $\mathcal{B}$ in time $q^{O(n)}$.

The next three subsections are devoted to the proofs of Propositions15 (Section5.3.3),16 (Section5.3.1), and17 (Section5.3.2). Note that, because the proof of Proposition15 is more complicated compared to the other two, we postpone it to the last.

Remark18. In fact, we expect that this construction works even for small finite fields. The bottleneck lies in Proposition15. If the probability $n^{O(1)}/q^{\Omega(1)}$ could be improved to $n^{O(1)}/q^{\Omega(n)}$, then we would be done. We believe it possible to utilize the structure of invariant subspaces under matrix actions over $\mathbb{F}_q$ to achieve this. However, we expect that the calculations will be tedious and heavy, so we hope to leave this to a future work.

5.3.1 Restricting to the diagonal group

Briefly speaking, conditions 1 and 2 ensure that we first restrict to monomial matrices. Conditions 3 and 4 prevent non-trivial permutations due to the following. As we assume $n = \Omega(1)$, by Observation12, $\sigma \in S_n$ either fixes 6 elements in $[n]$, or moves a set of 6 elements to another, disjoint, set of 6 elements. Condition 3 ensures that the second case could not happen. Condition 4 ensures that in the first case, the only possible invertible matrices that "preserves" the matrices G_i for $i \in P$ when multiplying from the left are scalar matrices. We now prove Proposition16.

Proof of Proposition16. Recall that we construct such $\widetilde{\mathtt{A}}$ and $\widetilde{\mathtt{B}}$ from $\mathtt{A}$ and $\mathtt{B}$, respectively, using the method in Section5.2. Let $\widetilde{\mathcal{A}}$ and $\widetilde{\mathcal{B}}$ be alternating matrix spaces in $\Lambda(7n, q)$, spanned by the frontal slices of $\widetilde{\mathtt{A}}$ and $\widetilde{\mathtt{B}}$, respectively. We want to show that $\widetilde{\mathcal{A}}$ and $\widetilde{\mathcal{B}}$ are isometric if and only if $\mathcal{A}$ and $\mathcal{B}$ are isometric via diagonal matrices. The if direction is straightforward. Suppose there exist $P = \mathrm{diag}(\alpha_1, \ldots, \alpha_n) \in \mathrm{diag}(n, q)$ and $Q \in \mathrm{GL}(m, q)$ such that $P^t \mathbf{A} P = \mathbf{B}^Q$. Let $\widetilde{P} = \begin{bmatrix} P & 0 \\ 0 & I_{6n} \end{bmatrix} \in \mathrm{GL}(7n, q)$. Let $\widetilde{Q} = \begin{bmatrix} Q & 0 \\ 0 & Q' \end{bmatrix} \in \mathrm{GL}(m + 4n^2)$, where $Q' = \mathrm{diag}(\alpha_1 I_{4n}, \ldots, \alpha_n I_{4n})$. Then it is easy to verify that $\widetilde{P}^t \widetilde{\mathtt{A}} \widetilde{P} = \widetilde{\mathtt{B}}^{\widetilde{Q}}$. Now we turn to the only if direction. If $\widetilde{\mathcal{A}}$ and $\widetilde{\mathcal{B}}$ are isometric, then there exists $\widetilde{P} \in \mathrm{GL}(7n, q)$ and $\widetilde{Q} \in \mathrm{GL}(m + 4n^2, q)$, such that $\widetilde{P}^t \widetilde{\mathtt{A}} \widetilde{P} = \widetilde{\mathtt{B}}^{\widetilde{Q}}$. Let $\widetilde{P} = \begin{bmatrix} P_{1,1} & P_{1,2} \\ P_{2,1} & P_{2,2} \end{bmatrix}$, where $P_{1,1}$ is of size $n \times n$. It can be checked easily, from the lateral viewpoint, that $P_{1,2} = 0$. As if not, then some H_i would appear in one of the last $6n$ lateral slices in $\widetilde{\mathtt{A}}^{\widetilde{P}}$. This would set this slice to be of rank $\geq 4n$ by

condition(1), which contradicts that the corresponding lateral slice of $\widetilde{\mathtt{B}}^{\widetilde{Q}}$ is of rank $\leq n$. It follows that $P_{1,1} \in \mathrm{GL}(n,q)$ and $P_{2,2} \in \mathrm{GL}(6n,q)$. We first claim that $P_{1,1}$ has to be a monomial matrix. If not, then one of the first n lateral slice of $\widetilde{\mathtt{A}}^{\widetilde{P}}$ has two distinct H_i and H_j. By condition(2), this slice is of rank $\geq 6n$, which contradicts that the corresponding lateral slice of $\widetilde{\mathtt{B}}^{\widetilde{Q}}$ is of rank $\leq 5n$. We further claim that $P_{1,1}$ has to be a diagonal matrix. If not, then suppose the non-trivial permutation underlying $P_{1,1}$ is $\sigma \in S_n$. Since we assumed $n = \Omega(1)$, by Observation12, one of the following two cases has to happen.

- $\exists\{i_1,\ldots,i_6\} \subseteq [n]$, $\{j_1,\ldots,j_6\} \subseteq [n]$, $|\{i_1,\ldots,i_6\} \cup \{j_1,\ldots,j_6\}| = 12$, such that $\sigma(i_k) = j_k$ for $k \in [6]$. We then claim the following.

  Claim19. For $\widetilde{P}^t\widetilde{\mathtt{A}}\widetilde{P} = \widetilde{\mathtt{B}}^{\widetilde{Q}}$ to hold, a necessary condition is that $\forall k \in [6]$, $P_{2,2}H_{j_k}$ and H_{i_k} have the same linear span.

  Proof. To see this, note that the i_kth lateral slice of $\widetilde{P}^t\widetilde{\mathtt{A}}\widetilde{P}$ is the j_kth lateral slice of $\widetilde{P}^t\widetilde{\mathtt{A}}$ (up to a scalar multiple). It is equal to the i_kth lateral slice of $\widetilde{\mathtt{B}}^{\widetilde{Q}}$. Then $\widetilde{P}^t$ acts on the left on the j_kth lateral slice of $\widetilde{\mathtt{A}}$. Noting that $P^t = \begin{bmatrix} P_{1,1}^t & P_{2,1}^t \\ 0 & P_{2,2}^t \end{bmatrix}$ and the j_kth lateral slice of $\widetilde{\mathtt{A}}$ is $C'_{j_k} = \begin{bmatrix} C_{j_k} & 0 \\ 0 & G_{j_k} \end{bmatrix}$, we see that $P^tC'_{j_k} = \begin{bmatrix} * & * \\ 0 & P_{2,2}^tG_{j_k} \end{bmatrix}$. (Here, C_i and G_i are defined in Section5.2.) On the other hand, we see that the i_kth lateral slice of $\widetilde{\mathtt{B}}^{\widetilde{Q}}$ is the i_kth lateral slice multiplied from the right by $\widetilde{Q}$. Our claim follows then by comparing the last $6n$ rows. □

  But the condition(3) excludes the existence of such $P_{2,2}$, so this cannot happen.

- $\exists\{i_1,\ldots,i_6\} \subseteq [n]$, i_k all different, such that $\sigma(i_k) = i_k$. In this case, for $\widetilde{P}^t\widetilde{\mathtt{A}}\widetilde{P} = \widetilde{\mathtt{B}}^{\widetilde{Q}}$ to hold, by the same argument as in the proof of Claim19, a necessary condition is that $P_{2,2}H_{i_k}$ and H_{i_k} have the same linear span. Then the condition(4) ensures that $P_{2,2} = \lambda I_{6n}$ for some $\lambda \neq 0 \in \mathbb{F}$ in this setting. Then because σ is non-trivial, σ moves some $i \in [n]$ to $j \in [n]$, $i \neq j$. By comparing the jth lateral slice of $\widetilde{P}^t\widetilde{\mathtt{A}}$ and the ith lateral slice of $\widetilde{\mathtt{B}}^{\widetilde{Q}}$, $P_{2,2}H_i = \lambda H_i$ and H_j have the same linear span, which is not possible because the condition(2) ensures that H_i and H_j span different subspaces.

We then have shown that $P_{1,1}$ must be a diagonal matrix. By comparing the top-left-front sub-tensors of size $n \times n \times m$ of $\widetilde{P}^t\widetilde{\mathtt{A}}\widetilde{P}$ and $\widetilde{\mathtt{B}}^{\widetilde{Q}}$, we arrive at the desired conclusion that $\mathcal{A}$ and $\mathcal{B}$ are isometric via the diagonal matrix $P_{1,1}$. □

5.3.2 Using the gadget for counting-to-decision reduction

The strategy follows closely the counting to decision reduction for graph isomorphism. We then apply the above strategy to get a counting to decision reduction for alternating matrix space isometry to prove Proposition17.

Proof of Proposition17. Our goal is to compute the number of isomorphisms from $\mathcal{A}$ to $\mathcal{B}$, where $\mathcal{A}, \mathcal{B} \leq \Lambda(n,q)$ are of dimension m. First, we use the decision oracle first to decide whether $\mathcal{A}$ and $\mathcal{B}$ are isometric. If not, the number of isometries is 0. If so, we need to caculate the order of the autometry group of $\mathcal{A}$, $\mathrm{Aut}(\mathcal{A})$. To do that, we first randomly sample n $6n \times 4n$ matrices $H_1, \ldots, H_n$ over $\mathbb{F}_q$, and verify whether they satisfy the four conditions in Section5.2 using Remark14. Note that this is where the algorithm needs to be a Las Vegas algorithm. Let $A = \mathrm{Aut}(\mathcal{A})$. Recall that e_i denotes the ith standard basis vector in $\mathbb{F}_q^n$. For $i \in [n]$, let $A_i = \{T \in A : \forall 1 \leq j \leq i, T(e_i) = \lambda_i e_i, \lambda_i \neq 0 \in \mathbb{F}_q\}$. Note that $A_n = A \cap \mathrm{diag}(n,q)$. We can calculate the order of A_n in time $q^{O(n)}$ by brute-force, i.e., enumerating all invertible diagonal matrices. Set $A_0 = A$. We then have the tower of subgroups $A_0 \geq A_1 \geq \cdots \geq A_n$. To compute the order of A_0, it is enough to compute $[A_i : A_{i+1}]$. Note that for $T, T' \in A_i$, $TA_{i+1} = T'A_{i+1}$ as left cosets in A_i if and only if $T(e_{i+1}) = \lambda T'(e_{i+1})$ for some $\lambda \neq 0 \in \mathbb{F}_q$. So $[A_i : A_{i+1}]$ is equal to the size of the orbit of e_{i+1} under A_i in the projective space. Let $v \in \mathbb{F}_q^n$. To test whether v is in the orbit of e_{i+1} under A_i in the projective space, we tranform $\mathcal{A}$ by $P^t \cdot P$, where $P \in \mathrm{GL}(n,q)$ sends e_{i+1} to v and e_j to e_j for $j \neq i+1$, to get $\mathcal{A}'$. We then add the diagonal restriction gadget to the first $i+1$ lateral slices and the first $i+1$ horizontal slices of $\mathcal{A}$ and $\mathcal{A}'$, to obtain $\widetilde{\mathcal{A}}$ and $\widetilde{\mathcal{A}}'$ respectively. Then feed $\mathcal{A}$ and $\mathcal{A}'$ to the decision oracle. By the functionality of the diagonal restriction gadget, v is in the orbit of e_{i+1} in the projective space if and only if $\widetilde{\mathcal{A}}$ and $\widetilde{\mathcal{A}}'$ are isometric. Enumerating $v \in \mathbb{F}_q^n$ up to scalar multiples gives us the size of the orbit of e_{i+1} under A_i in the projective space. This finishes the description of the algorithm. A small caveat in the above is that our gadget requires $n = \Omega(1)$, so we cannot start from A_0 at the beginning. This issue can be revolved by noting that the order of A_c, for any constant c, can be computed in time $q^{O(n)}$, by enumerating all possible images of $e_1, \ldots, e_c$ in time $q^{O(n)}$, adding the diagonal restriction gadget, and utilizing the decision oracle. □

5.3.3 Random H_i's satisfy the requirements when $q = n^{\Omega(1)}$

In the following we will encounter random matrices over $\mathbb{F}_q$ as well as random subspaces in $\mathbb{F}_q^n$. There is a subtle point which we want to clarify now. Let $m \leq n$. Note that there are $\binom{n}{m}_q$ subspaces of $\mathbb{F}_q^n$, and there are $N_1 = (q^n - 1) \cdots (q^n - q^{m-1})$ rank-m matrices of size $n \times m$. It can be seen easily that each m-dimensional subspace V of $\mathbb{F}_q^n$ has $N_2 = (q^m - 1) \cdots (q^m - q^{m-1})$ many representations as rank-m matrices of size $n \times m$, i.e. the columns of the matrix span V. It follows that we can work with random rank-m matrices of size $n \times m$ as if we are working with random m-dimensional subspaces of $\mathbb{F}_q^n$. Such correspondences will be used implicitly for other structures, including direct sum decompositions. Now let us get back to our question. We shall show that a random choice of H_i, $i \in [n]$, would satisfy the four conditions we imposed on H_i's. We will prove that for conditions $k = 1, 2, 3$,

$$\Pr[\text{random } H_i \text{ not satisfy condition } k] \leq \frac{n^{O(1)}}{q^{\Omega(n)}}.$$

Once these hold, by a union bound, we have

$$\Pr[\exists i \in [3],\ \text{random } H_i \text{ not satisfy condition } i] \leq \frac{n^{O(1)}}{q^{\Omega(n)}}.$$

For condition(4), we will prove that

$$\Pr[\text{random } H_i \text{ not satisfy condition 4} \mid H_i \text{ satisfy conditions 1,2,3}] \leq \frac{n^{O(1)}}{q^{\Omega(1)}}.$$

This then would allow us to conclude that when $q = n^{\Omega(1)}$, random H_i's satisfy all the four conditions. We examine the first three conditions one by one.

1. For condition(1), by Fact13(2), we have $\Pr[\exists i \in [n], \mathrm{rk}(H_i) < 4n] \leq n \cdot \Pr[\mathrm{rk}(H_i) < 4n] \leq 2n/q^{2n+1}$.

2. For condition(2), noting that the block matrix $(H_i\ H_j)$ is a random $6n \times 8n$ matrix over $\mathbb{F}_q$, by Fact13(2), we have

$$\Pr[\exists i \neq j \in [n], \mathrm{rk}((H_i\ H_j)) < 6n] \leq \binom{n}{2} \cdot \frac{2}{q^{8n-6n+1}} \leq \frac{n^2}{q^{2n+1}}.$$

3. For condition(3), let $I = (H_{i_1}\ \cdots\ H_{i_6})$, and $J = (H_{j_1}\ \cdots\ H_{j_6})$. We see that $\mathcal{C}$ is non-empty if and only if there exists $L \in \mathrm{GL}(6n, q)$ and $R_k \in \mathrm{GL}(4n, q)$, $k \in [6]$, such that $LH_{i_k}R_k = H_{j_k}$. Note that the orbit of I under this group action is of size at most $q^{(6n)^2+6\cdot(4n)^2} = q^{132n^2}$. Since i_k and j_ℓ are all different, the probability of J belonging to this orbit is $\leq q^{132n^2}/q^{144n^2} = 1/q^{12n^2}$. We then have

$$\Pr[\exists i_k, j_k \in [n], k \in [6], i_k, j_k \text{ all different}, \mathcal{C} = \emptyset] \leq \binom{n}{12} \frac{2}{q^{12n^2}} \leq \frac{n^{12}}{q^{12n^2}}.$$

We now focus on condition(4). For condition(4), we first assume that the conditions(1) and(2) as above hold. Then V_i's are random $4n$-dimensional subspaces of $\mathbb{F}_q^{6n}$. Note that

$$\Pr[\exists i_k \in [n], k \in [6], i_k \text{ all different}, \mathcal{S} \text{ non-scalar}] \leq n^6 \cdot \Pr[\mathcal{S} \text{ non-scalar for } V_1, \ldots, V_6].$$

So we turn to study $\Pr[\mathcal{S}$ non-scalar for $V_1, \ldots, V_6]$, and will show that it is $\leq 1/q^{\Omega(1)}$. Let $U_1 = V_1 \cap V_2$, $U_2 = V_2 \cap V_3$, and $U_3 = V_1 \cap V_3$. Let $W_1 = V_4 \cap V_5$, $W_2 = V_5 \cap V_6$, and $W_3 = V_4 \cap V_6$. Since conditions(1) and(2) hold, we have $\dim(U_i) = \dim(W_i) = 2n$. We claim that with probability $\geq 1 - 2/q$, $\mathbb{F}_q^{6n} = U_1 \oplus U_2 \oplus U_3$, i.e., $U_1 \cup U_2 \cup U_3$ span $\mathbb{F}_q^{6n}$. This can be seen as follows. Since we assumed conditions(1) and(2), this happens if and only if $V_1 \cap V_2$ and V_3 together span $\mathbb{F}_q^{6n}$. Therefore we calculate, using Fact13(1),(3), and(5), that

$$\begin{aligned}
&\Pr[V_3 \text{ is a complement subspace of } V_1 \cap V_2] \\
&= q^{2n\cdot 4n} \Big/ \binom{6n}{4n}_q = \frac{(q^{6n} - q^{2n})(q^{6n} - q^{2n+1}) \cdots (q^{6n} - q^{6n-1})}{(q^{6n} - 1)(q^{6n} - q) \cdots (q^{6n} - q^{4n-1})} \\
&\geq \frac{(q^{6n} - q^{2n})(q^{6n} - q^{2n+1}) \cdots (q^{6n} - q^{6n-1})}{q^{6n} \cdot q^{6n} \cdots q^{6n}}
\end{aligned}$$

$$= (1-1/q^{4n})(1-1/q^{4n-1})\cdots(1-1/q) \geq 1-\sum_{i=1}^{4n} 1/q^i \geq 1-2/q.$$

It follows that with probability $\geq 1-4/q$, we can assume in addition that W_i form a direct sum decomposition of $\mathbb{F}_q^{6n}$. Therefore, we turn to bound the probability that there exists a non-scalar invertible matrix stabilizing these two direct sum decompositions of $\mathbb{F}_q^{6n}$. Since i_k are all different, the two direct sum decompositions $U_1 \oplus U_2 \oplus U_3$ and $W_1 \oplus W_2 \oplus W_3$ are independent. So we can assume that U_i is spanned by those standard basis vectors $\vec{e}_{2n(i-1)+1}, \ldots, \vec{e}_{2ni}$, $i = 1, 2, 3$. The group that stabilizes this direct sum decomposition $U_1 \oplus U_2 \oplus U_3$ consists of $\begin{bmatrix} D_1 & 0 & 0 \\ 0 & D_2 & 0 \\ 0 & 0 & D_3 \end{bmatrix} \in \mathrm{GL}(6n, \mathbb{F}_q)$ where D_i is of size $2n \times 2n$. The question then becomes to bound the probability for a random $W_1 \oplus W_2 \oplus W_3$ to be stabilized by a non-scalar matrix of the above form. This can be formulated as the following linear algebraic problem. (Recall the correspondence between random m-dimensional subspaces and random rank-m matrices as discussed at the beginning of the subsection.) Let $W = \begin{bmatrix} W_{11} & W_{12} & W_{13} \\ W_{21} & W_{22} & W_{23} \\ W_{31} & W_{32} & W_{33} \end{bmatrix} \in \mathrm{GL}(6n, q)$ be a block matrix where W_{ij} is of size $2n \times 2n$. Suppose the columns of $\begin{bmatrix} W_{1i} \\ W_{2i} \\ W_{3i} \end{bmatrix}$ span W_i. Then $D = \mathrm{diag}(D_1, D_2, D_3)$ stabilizes $W_1 \oplus W_2 \oplus W_3$ if and only if there exists a block diagonal matrix $E = \mathrm{diag}(E_1, E_2, E_3)$, $E_i \in \mathrm{GL}(2n, q)$, such that

$$\begin{bmatrix} D_1 & 0 & 0 \\ 0 & D_2 & 0 \\ 0 & 0 & D_3 \end{bmatrix} \begin{bmatrix} W_{11} & W_{12} & W_{13} \\ W_{21} & W_{22} & W_{23} \\ W_{31} & W_{32} & W_{33} \end{bmatrix} = \begin{bmatrix} W_{11} & W_{12} & W_{13} \\ W_{21} & W_{22} & W_{23} \\ W_{31} & W_{32} & W_{33} \end{bmatrix} \begin{bmatrix} E_1 & 0 & 0 \\ 0 & E_2 & 0 \\ 0 & 0 & E_3 \end{bmatrix}. \tag{5}$$

Note that each direct sum decomposition $W_1 \oplus W_2 \oplus W_3$, $\dim(W_i) = 2n$, has $6 \cdot |\mathrm{GL}(2n, q)|^3$ such matrix representations. (The factor 6 takes care of the orders of the three summands.) So the question becomes to bound the probability for a random invertible matrix to have a non-scalar D and E satisfying Equation5. First, note that Equation5 holds if and only if $D_i W_{i,j} = W_{i,j} E_j$ for $i, j \in [3]$.

Claim20. When $q = \Omega(1)$, we have $\Pr[\forall i, j \in [3], \mathrm{rk}(W_{i,j}) = 2n] \geq 1 - 20/q$.

Proof. Let us work in the setting when W is a random matrix, not necessarily the one above. Then $\Pr[\mathrm{rk}(W) = 6n] \geq 1-2/q$. For any $i, j \in [3]$, $\Pr[\mathrm{rk}(W_{i,j}) < 2n] \leq 2/q$, so $\Pr[\exists i, j \in [3], \mathrm{rk}(W_{i,j}) < 2n] \leq 18/q$. It follows that

$$\begin{aligned} &\Pr[\exists i, j \in [3], \mathrm{rk}(W_{i,j}) < 2n \mid \mathrm{rk}(W) = 6n] \\ &= \frac{\Pr[\exists i, j \in [3], \mathrm{rk}(W_{i,j}) < 2n \wedge \mathrm{rk}(W) = 6n]}{\Pr[\mathrm{rk}(W) = 6n]} \\ &\leq \frac{18/q}{1-2/q} = \frac{18}{q-2} \leq \frac{20}{q}, \end{aligned}$$

where the last inequality uses that $q = \Omega(1)$. □

So we assume that $\mathrm{rk}(W_{i,j}) = 2n$ for all $i, j \in [3]$ in the following, with a loss of probability $\leq 20/q$. For $i \in [3]$, by $D_i W_{ii} = W_{ii} E_i$, we have $D_i = W_{ii} E_i W_{ii}^{-1}$. For $i \neq j$, by $(W_{jj} E_j W_{jj}^{-1}) W_{ji} = D_j W_{ji} = W_{ji} E_i$, we have $E_j = W_{jj}^{-1} W_{ji} E_i W_{ji}^{-1} W_{jj}$. Again for $i \neq j$, we have $W_{ii} E_i W_{ii}^{-1} W_{ij} = D_i W_{ij} = W_{ij} E_j = W_{ij} W_{jj}^{-1} W_{ji} E_i W_{ji}^{-1} W_{jj}$. It follows that

$$\forall i, j \in [3], i \neq j, \quad E_i W_{ii}^{-1} W_{ij} W_{jj}^{-1} W_{ji} = W_{ii}^{-1} W_{ij} W_{jj}^{-1} W_{ji} E_i.$$

In particular, E_3 commutes with $X = W_{33}^{-1} W_{32} W_{22}^{-1} W_{23}$ and $Y = W_{33}^{-1} W_{31} W_{11}^{-1} W_{13}$. Since W_{ij} are independent random invertible matrices, X and Y are independent random invertible matrices. We now resort to the following classical result.

Theorem21 ([41], cf. also [23,40]). Let X and Y be two random matrices in $\mathrm{SL}(n, q)$. Then the probability of X and Y not generating $\mathrm{SL}(n, q)$ is $\leq 1/q^{\Omega(n)}$.

Back to our setting, the above theorem implies that the group G generated by random X and Y from $\mathrm{GL}(2n, q)$ contains $\mathrm{SL}(2n, q)$ with probability $\geq 1 - 1/q^{\Omega(n)}$. It follows that E_3 belongs to the centralizer of G, so E_3 must be a scalar matrix. Then note that D_i's and other E_i's are all conjugates of E_3. So we have $\forall i \in [3], D_i = E_i = \lambda I_{2n}$ for some $\lambda \neq 0 \in \mathbb{F}_q$. Summarizing the above, we have

$$\begin{aligned}
&\Pr[\mathcal{S} \text{ non-scalar for } V_1, \ldots, V_6] \\
&\leq \Pr[\mathcal{S} \text{ non-scalar for } V_i \wedge \mathbb{F}_q^{6n} = U_1 \oplus U_2 \oplus U_3 = W_1 \oplus W_2 \oplus W_3] + 4/q \\
&\leq \Pr[\mathcal{S} \text{ non-scalar for } V_i \mid \mathbb{F}_q^{6n} = U_1 \oplus U_2 \oplus U_3 = W_1 \oplus W_2 \oplus W_3] + 4/q \\
&\leq \Pr[D \text{ non-scalar for } W \wedge \forall i, j \in [3], \mathrm{rk}(W_{ij}) = 2n] + 20/q + 4/q \\
&\leq \Pr[D \text{ non-scalar for } W \mid \forall i, j \in [3], \mathrm{rk}(W_{ij}) = 2n] + 24/q \\
&\leq 1/q^{\Omega(n)} + 24/q \leq 1/q^{\Omega(1)},
\end{aligned}$$

when $q = n^{\Omega(1)}$. This concludes the proof of Proposition15. □

6.2 Constructive Baer Correspondence and Theorems A and B

Let us review Baer's Correspondence [7], which connects alternating bilinear maps with p-groups of class 2 and exponent p. Let P be a p-group of class 2 and exponent p, $p > 2$. Suppose the commutator subgroup $[P, P] \cong \mathbb{Z}_p^m$ and $P/[P, P] \cong \mathbb{Z}_p^n$. Then the commutator map $[,] : P/[P, P] \times P/[P, P] \to [P, P]$ is an alternating bilinear map. Conversely, let $\phi : \mathbb{Z}_p^n \times \mathbb{Z}_p^n \to \mathbb{Z}_p^m$ be an alternating bilinear map. Then a p-group of class 2 and exponent p, denoted as P_ϕ can be defined as follows. The group elements are from $\mathbb{Z}_p^n \times \mathbb{Z}_p^m$, and the group product $\cdot$ is defined as

$$(u, v) \cdot (u', v') = (u + u', v + v' + \tfrac{1}{2}\phi(u, u')).$$

We say that $(A, B) \in \mathrm{GL}(n, p) \times \mathrm{GL}(m, p)$ is a pseudo-autometry of ϕ, if $\phi = B \circ \phi \circ A$. Wilson [71] elucidated the structure of $\mathrm{Aut}(P_\phi)$ in terms of the pseudo-autometry group of ϕ, that we denote $\Psi\,\mathrm{Aut}(\phi)$. Here we recall the consequence of Wilson's result that we need for counting group isomorphisms.

Proposition23 (Wilson [71, Prop.3.8], see [15, Prop.2.4] for notation closer to ours). For $\phi : \mathbb{Z}_p^n \times \mathbb{Z}_p^n \to \mathbb{Z}_p^m$ an alternating bilinear map,

$$|\operatorname{Aut}(P_\phi)| = |\Psi \operatorname{Aut}(\phi)| p^{nm},$$

where $\Psi \operatorname{Aut}(\phi)$ denotes the pseudo-autometry group of ϕ.

We then state a lemma which can be viewed as a constructive version of Baer's Correspondence, communicated to us by James B. Wilson.

Lemma24 (Constructive version of Baer's Correspondence for matrix groups). Let p be an odd prime. Over the finite field $\mathbb{F} = \mathbb{F}_{p^e}$, Alternating Matrix Space Isometry is equivalent to Group Isomorphism for matrix groups over $\mathbb{F}$ that are p-groups of class 2 and exponent p. More precisely, there are functions computable in time $\operatorname{poly}(n, m, \log|\mathbb{F}|)$:

- $G : \Lambda(n, \mathbb{F})^m \to \mathrm{M}(n + m + 1, \mathbb{F})^{n+m}$ and
- $\operatorname{Alt} : \mathrm{M}(n, \mathbb{F})^m \to \Lambda(m, \mathbb{F})^{O(m^2)}$

such that: (1) for an alternating bilinear map $\mathbf{A}$, the group generated by $G(\mathbf{A})$ is the Baer group corresponding to $\mathbf{A}$, (2) G and Alt are mutually inverse, in the sense that the group generated by $G(\operatorname{Alt}(M_1, \ldots, M_m))$ is isomorphic to the group generated by $M_1, \ldots, M_m$, and conversely $\operatorname{Alt}(G(\mathbf{A}))$ is pseudo-isometric to $\mathbf{A}$.

Proof. First, let G be a p-group of class 2 and exponent p given by m generating matrices of size $n \times n$ over $\mathbb{F}$. Then from the generating matrices of G, we first compute a generating set of $[G, G]$, by just computing all the commutators of the given generators. We can then remove those redundant elements from this generating set in time $\operatorname{poly}(\log|[G, G]|, \log|\mathbb{F}|)$, using Luks' result on computing with solvable matrix groups [51]. We then compute a set of representatives of a non-redundant generating set of $G/[G, G]$, again using Luks's aforementioned result. From these data we can compute an alternating bilinear map representing the commutator map of G in time $\operatorname{poly}(n, m, \log|\mathbb{F}|)$. Conversely, let an alternating bilinear map be given by $\mathbf{A} = (A_1, \ldots, A_m) \in \Lambda(n, \mathbb{F})^m$. From $\mathbf{A}$, for $i \in [n]$, construct $B_i = [A_1 e_i, \ldots, A_m e_i] \in \mathrm{M}(n \times m, \mathbb{F})$, where e_i is the ith standard basis vector of $\mathbb{F}^n$. That is, the jth column of B_i is the ith column of A_j. Then for $i \in [n]$, construct

$$\widetilde{B}_i = \begin{bmatrix} 1 & e_i^t & 0 \\ 0 & I_n & B_i \\ 0 & 0 & I_m \end{bmatrix} \in \mathrm{GL}(1 + n + m, \mathbb{F}),$$

where $e_i \in \mathbb{F}^n$, and for $j \in [m]$, construct

$$\widetilde{C}_j = \begin{bmatrix} 1 & 0 & e_j^t \\ 0 & I_n & 0 \\ 0 & 0 & I_m \end{bmatrix} \in \mathrm{GL}(1 + n + m, \mathbb{F}),$$

where $e_j \in \mathbb{F}^m$. Let $G(\mathbf{A})$ be the matrix group generated by $\widetilde{B}_i$ and $\widetilde{C}_j$. Then it can be verified easily that, $G(\mathbf{A})$ is isomorphic to the Baer group corresponding to the alternating bilinear map defined by $\mathbf{A}$. In particular, $[G, G] \cong \mathbb{F}^m \cong \mathbb{Z}_p^{em}$ (isomorphism of abelian groups), and $G/[G, G] \cong \mathbb{F}^n \cong \mathbb{Z}_p^{en}$. This construction can be done in time $\operatorname{poly}(n, m, \log|\mathbb{F}|)$. □

Proof of TheoremB. For the counting-to-decision reduction, we basically follow the above routine, but with a twist, because of the minor distinction between alternating matrix space isometry, and alternating bilinear map pseudo-isometry. Let us briefly explain this issue. Suppose from an alternating bilinear map $\phi : \mathbb{Z}_p^n \times \mathbb{Z}_p^n \to \mathbb{Z}_p^m$ we constructed the p-group P_ϕ of class 2 and exponent p; by Proposition23 $|\operatorname{Aut}(P_\phi)| = p^{nm}|\Psi \operatorname{Aut}(\phi)|$, so by multiplying the result by p^{nm}, it is necessary and sufficient to count the psuedo-autometries of ϕ. Towards that end, let $(C_1, \ldots, C_m) \in \Lambda(n, p)$ be a matrix representation of ϕ. If C_i's are linearly independent, then for a pseudo-autometry $(A, B) \in \mathrm{GL}(n, p) \times \mathrm{GL}(m, p)$, given A there exists a unique B that makes (A, B) a pseudo-autometry. If C_i's are not linearly independent, say the linear span of C_i's is of dimension m', then the number of B such that (A, B) is a pseudo-autometry (assuming there are any) is $|\mathrm{M}((m - m') \times m', p)||\mathrm{GL}(m - m', p)| = p^{m'(m-m')}|\mathrm{GL}(m - m', p)|$. To see this, suppose that we have taken linear combinations of the C_i so that $C_1, \ldots, C_{m'}$ are linearly independent and $C_{m'+1}, C_{m'+2}, \ldots, C_m$ are zero. Then without changing the C_i, we may take any invertible linear combination among $C_{m'+1}, \ldots, C_m$ (a copy of $\mathrm{GL}(m - m', p)$), and we may add any linear combination of the last $m - m'$ matrices to the first m' matrices (a copy of $\mathrm{M}((m - m') \times m', p)$). The counting to decision reduction for Alternating Matrix Space Isometry computes the number of $A \in \mathrm{GL}(n, p)$ so that there exists some $B \in \mathrm{GL}(m, p)$ such that (A, B) is a pseudo-autometry. So it needs to be multiplied by a factor of $p^{m'(m-m')}|\mathrm{GL}(m - m', p)|$. □

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